26-BS-M101 — Mathematics – I

Course Code	26-BS-M101	**Category**	Basic Science Course
Course Title	Mathematics – I	**L – T – P**	3 – 0 – 0
Credits	3	**Contact Hours**	36
Semester	First	**Assessment**	Theory: CA 30 + ESE 70

Prerequisites

The course builds on the Higher Secondary Mathematics of WBCHSE, CBSE or ISC. Being a first-semester course it is scoped a little lighter, since admission to the first year is habitually late and the effective teaching window is shorter than the calendar suggests. The following are assumed:

- Differentiation and integration of elementary functions; the idea of a limit and of continuity (CBSE XII Units III–IV; WBCHSE Class XII; ISC).
- Matrices and determinants to the third order and the solution of a linear system (CBSE XII Unit III).
- Vectors in two and three dimensions (CBSE XII Unit IV) and complex numbers in algebraic and polar form (CBSE XI).
- A ten-hour bridge is conducted for the whole class in the first three weeks, re-establishing differentiation, integration and the standard notation; it is recorded in the whole-semester component and is not separately credited.

Course Objectives

- To establish the differential and integral calculus of one and several variables as the working language of engineering analysis.
- To develop competence in the matrix and eigenvalue methods that underlie later courses across every branch.
- To introduce the convergence of sequences and series as the basis of approximation.
- To do so on a common, branch-neutral footing, leaving discipline-specific mathematics to Semester III.

Course Outcomes

On completion of the course the student will be able to:

CO	Statement	Bloom
CO1	Apply the differential calculus of one and several variables, including partial differentiation, to engineering functions.	L3 – Apply
CO2	Evaluate definite, improper and multiple integrals and use them to compute area, volume and average value.	L3 – Apply
CO3	Apply matrix methods — rank, the solution of linear systems, eigenvalues and eigenvectors — to engineering problems.	L3 – Apply
CO4	Test an infinite series for convergence and construct elementary power-series approximations.	L4 – Analyse

CO–PO Articulation Matrix

Correlation: 3 substantial, 2 moderate, 1 slight, – none. Mapped against the eleven Programme Outcomes of the NBA (GAPC v4.0).

CO \ PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	1	1	1	–	–	–	–	–	1
CO2	3	2	1	1	1	–	–	–	–	–	1
CO3	3	3	2	1	2	–	–	–	–	–	1
CO4	3	2	1	1	1	–	–	–	–	–	2

Course Plan — Module-wise Syllabus

Module	Title	Contents	Hours
I	Calculus of a Single Variable	Review in brief of limits, continuity and differentiation; successive differentiation and Leibnitz's theorem; Rolle's and the mean-value theorems; Taylor's and Maclaurin's series with elementary applications; maxima and minima of a function of one variable; definite and improper integrals and the reduction formulae in outline.	9
II	Calculus of Several Variables	Functions of two and more variables; partial derivatives; the total differential and its use in error estimation; the chain rule; Euler's theorem on homogeneous functions; maxima and minima of a function of two variables and the method of Lagrange multipliers in outline.	9
III	Multiple Integrals	Double integrals in Cartesian and polar coordinates; change of order of integration; computation of area and volume; the idea of the triple integral.	8
IV	Matrices and Linear Algebra	Rank of a matrix by elementary transformation; consistency and solution of a linear system; eigenvalues and eigenvectors; the Cayley–Hamilton theorem and its use; diagonalisation in outline. The linear-algebra content is generic; discipline-specific extensions are left to Mathematics-III.	10

Cross-cutting notes. This is a generic mathematics course, common to every branch of the University; it deliberately carries no discipline-specific material. Where a programme requires mathematics particular to it — discrete structures and graph theory for the computing branches, additional transforms and complex analysis for the electronics and electrical branches, additional statistics and numerical methods for the others — that requirement is to be met either in Mathematics-III in Semester III, which each Department frames for its own programme, or in Module I of the relevant discipline course where it is small enough to be carried there. The first-year mathematics is thus kept uniform, so that the common first-year examination and inter-branch mobility are preserved.

Text and Reference Books

- Reena Garg, Mathematics - I, Calculus and Linear Algebra, Khanna Publishing House (AICTE Prescribed Textbook)
- Reena Garg, Advanced Engineering Mathematics, Khanna Publishing House
- Erwin Kreyszig, Advanced Engineering Mathematics, Wiley.
- G. B. Thomas and R. L. Finney, Calculus and Analytic Geometry, Pearson.
- B. S. Grewal, Higher Engineering Mathematics, Khanna Publishers.
- N. P. Bali and M. Goyal, A Textbook of Engineering Mathematics, Laxmi Publications.

Suggested SWAYAM / NPTEL and Online Resources

- NPTEL — Engineering Mathematics I (Prof. Jitendra Kumar, IIT Kharagpur), 12 weeks.
- NPTEL — Multivariable Calculus, 12 weeks.
- NPTEL — Matrix Analysis with Applications / Linear Algebra, 12 weeks.
- SWAYAM — Mathematics I (AICTE / SWAYAM), as offered.

Question-paper note. The paper is predominantly a problem paper: at least two-thirds of the marks of Groups B and C are for the working of problems, not for statement of theory. No question draws on discipline-specific material, since the class is common to all branches.